Time Series Anomaly Detection in Vehicle Sensors Using Self-Attention Mechanisms

Ze Zhang, Yue Yao, Windo Hutabarat, Michael Farnsworth, Divya Tiwari, and Ashutosh Tiwari

***Abstract*— Connected autonomous vehicles (CAVs) offer significant enhancements in coordinated traffic and safety through real-time vehicle-to-vehicle or vehicle-to-infrastructure communications, establishing them as a potent tool for augmenting driving tasks. However, the extensive information-sharing framework inherent in CAVs amplifies the risk associated with sensor anomalies, posing challenges to the reliability and security of the system. Responding to this timely research challenge, this study proposes a novel anomaly detection method, namely Dual-channel Self-attention-based Convolutional Neural Network (DSA-CNN) for multivariate time series data. Through the introduction of the Dual-channel Self-attention Mechanism, DSA-CNN can progressively and autonomously extract spatiotemporal features from multivariate time series data. The proposed method was tested under a variety of common threatening sensor anomaly patterns of CAVs summarised in the literature, and evaluated under multiple different performance metrics. The results demonstrate its advantages in detecting minor anomalies and enhancing sensitivity, outperforming previously reported methods in the literature. Across all experimental scenarios, an average sensitivity improvement of 2.53% was observed, complemented by an average F1 score increase of 1.47%. In CAV settings, maintaining high sensitivity to ensure fewer undetected anomalies, alongside the ability to detect small anomalies, can be more important for the robustness and safety measures of CAV systems.**

***Index Terms*— Anomaly detection, connected and autonomous vehicles (CAVs), deep learning, attention mechanism, multi-sensor system, intelligent transportation system (ITS).**

I. Introduction

CONNECTED and Automated Vehicles (CAVs), reflecting a convergence of automation and connectivity advancements, are receiving a surge in interest from academia, industry, and governments. For example, the CAVs are anticipated by the UK government to create a multi-billion pound industry [1] and ranked top 10 in KPMG's Autonomous Vehicles Readiness Index 2020 [2]. The center of this paradigm can be the incorporation of a variety of technologies, such as advanced sensors, vehicle-to-vehicle (V2V) communications, vehicle-to-infrastructure (V2I) communications, and Artificial Intelligence (AI) for autonomous navigation and decision-making [3]. By leveraging connectivity, real-time traffic information such as changing road conditions, unforeseen events, and safety warnings can be shared among vehicles, enabling the global optimisation of traffic flow, reduced congestion, and enhanced road safety [4], [5]. Therefore, compared with conventional autonomous vehicles, CAVs offer considerable advantages to the future of transportation.

Due to the extensive information-sharing framework inherent in CAVs, the reliability of sensors, which can be intrinsically tied to the safety and security of CAVs, has emerged as a matter of concern in recent years [2]. However, given a large amount of data transmission [6], cyber-attacks, and the highly complex operating environment of the vehicles, sensor anomalies which refer to where sensor readings fail to accurately represent the actual physical processes they are monitoring can be difficult to avoid and pose a significant threat to the core functions of CAVs, such as autopilot control system and traffic coordination [6]. As a result, sensor anomaly detection for CAVs has become a research focus in recent years.

The aim of sensor anomaly detection is to identify the readings that deviate from the expected values, which can result from many reasons, such as sensor malfunction and cyber-attacks [7]. By doing so, timely intervention can be enabled to avoid potentially severe consequences. Anomaly detection has been extensively researched in the literature and commonly used methods include, but are not limited to, statistical methods, signal processing, time series analysis (TSA), and data-driven methods, especially deep learning. In a statistical method, sensor data can be modelled by a statistical model and readings that deviate from the model prediction can be considered anomalous according to a pre-defined threshold [8]. The statistical models for this type of method can either be built from domain knowledge or statistically derived from the data collected, such as the Gaussian model [9], histogram-based model [10], or Hypothesis testing method [11]. Generally speaking, most of them can be interpretable and computationally efficient. However, this type of method requires that the sensor data can be characterised by a specific distribution, which may not be the case for many applications, especially for data of high dimensionality [12]. Signal processing-based anomaly detection, such as the Fourier transform and the wavelet transform is another commonly used method, which can be used for a variety of sensor data, such as acoustic,

Manuscript received 2 June 2023; revised 11 November 2023 and 24 March 2024; accepted 20 May 2024. Date of publication 2 September 2024; date of current version 1 November 2024. This work was supported in part by the Engineering and Physical Sciences Research Council (EPSRC) through the Made Smarter Innovation–Research Centre for Connected Factories under Grant EP/V062123/1, and in part by the Royal Academy of Engineering (RAEng) and Airbus under the Research Chairs and Senior Research Fellowships Scheme under Grant RCSRF1718/5/41. The Associate Editor for this article was F. Xia. *(Corresponding author: Ze Zhang.)*

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Digital Object Identifier 10.1109/TITS.2024.3415435

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vision, and physical parameters data. In [13], a wavelet-based method was proposed to detect anomalies in the presence of noise, and they achieved remarkable performance with respect to detection rate and computational efficiency. As signal processing methods attempt to form unique descriptions of different signals, they often have an advantage when dealing with unseen anomalies. However, similar to statistical methods, their performance may heavily depend on making assumptions (e.g. quasi-stationarity) to processes such as noise distribution, which may limit their applications [12]. In terms of time-series analysis-based anomaly detection, it can also be a prevailing methodology [14], such as Kalman filtering [15], autoregressive moving average [16], symbolic TSA [17]. However, according to [12], this type of method may suffer from degrading performance when the anomaly causes dramatic changes to the original sensor readings.

Deep learning as a promising option for sensor anomaly detection has received more attention recently, particularly in scenarios where there are large amounts of data with complex patterns and high dimensionality [18], [19]. This is because deep learning methods typically can learn feature representations of the data that contain complex patterns and dependencies automatically. Compared with conventional methods, deep learning-based methods normally do not require any assumptions on the distribution of input data or noise, making it applicable in most areas with a certain amount of data. Convolutional Neural Network (CNN) can be one of the most commonly used methods and it has been shown to be effective in sensor anomaly detection [20], especially for multivariate inputs. For example, in [21], Chen et al. proposed a CNN-based anomaly detection algorithm for multiple sensor streams. However, CNN can be limited in its ability to model global features due to its reliance on a large number of local kernels to extract features, resulting in difficulty in capturing long-range dependencies, such as inter-sensor dependencies and time dependencies. The above shortcomings of CNNs can be well compensated by the Recurrent Neural Networks (RNNs), which model time dependence through a recurrent structure. The Long Short-Term Memory (LSTM) can be one of the representative RNNs. In [22], LSTM and CNN were combined to detect sensor anomalies for CAVs. They achieved a state-of-the-art performance on a public CAV dataset. However, since LSTM can be an RNN structure, despite the advantage over CNNs of introducing time-dependent modelling, the vanishing gradient may be non-negligible, which inherently limits its ability to retain information over long sequences [23], resulting in small anomalies in the time series being difficult to detect. Therefore, this work proposes a Dual-channel Self-attention-based CNN (DSA-CNN) to mitigate the limitations of both CNNs and RNNs in sensor anomaly detection, aiming to enhance the detection of small anomalies and reduce the incidence of undetected faults.

Transformer architectures have shown great potential for anomaly detection [24], [25]. In the self-attention mechanism of the Transformer method, data from all time points can be processed in parallel, instead of the recurrent computation in RNNs, making it significantly better than RNNs in capturing long-range dependence without suffering from vanishing gradient problem [26], [27]. Due to the advantages of CNN in local feature extraction, our proposed DSA-CNN combines two different deep learning building blocks, self-attention, and CNN, to improve its performance for anomaly detection in multi-sensor time series, especially for small anomalies. The main contributions can be summarised as follows:

- A new method, the DSA-CNN, has been developed, incorporating the self-attention mechanism specifically for detecting anomalies in multivariate time series sensor data. This model effectively identifies anomalous readings within the sensor data streams over a given time window.
- In our proposed method, the spatiotemporal features which can be vital for anomaly detection [28] can be extracted and integrated progressively and automatically with the help of separate attention channels during the learning process, eliminating the artificial signal processing stage for extracting the spatiotemporal features before designing an anomaly detection algorithm.
- This study has achieved a notable incremental improvement in performance compared to existing methods applied to the dataset used by this study. The experiment results show that our proposed DSA-CNN has a clear advantage in handling small anomalies, and improves the sensitivity significantly among all the experimental conditions being evaluated. This means our proposed method has fewer undetected anomalies which can be harmful in real systems compared with the other methods in the literature.

The rest of the paper is organised as follows. Section II formulates the problem being worked on and describes the dataset being used in this paper. Section III provides the details of our methodology, including the model architecture design and computation flow. Section IV provides the experiment settings and results comparison of the proposed method. A comprehensive discussion as well as the drawbacks of the proposed method can be found in Section V before Section VI where the conclusion is presented.

II. Problem Statement and Dataset Description

The concept of sensor anomaly refers to the situation in which sensor readings deviate from their expected values by producing faulty data and failing to reflect the actual physical processes, causing potential errors in decisions that depend on these readings. Addressing sensor anomaly, as a subset of Corner Cases, can be a necessary part of the consideration in the design of autonomous systems [29]. This work focuses on developing a supervised learning-based method to identify anomalous data in multi-sensor time-series signals to avoid serious consequences. The output of our proposed model can be the anomaly conditions of the current time window, hence we formulated the anomaly detection problem as a multivariate time series classification (TSC) problem. According to [30], [31], [32], and [33], in CAV settings, the commonly observed sensor abnormal patterns caused by cyber-attacks and sensor malfunctions can be summarised by the 4 types, namely Instant, Constant, Gradual drift, and Bias. A detailed description of the above-mentioned patterns in CAVs can be found in [15].

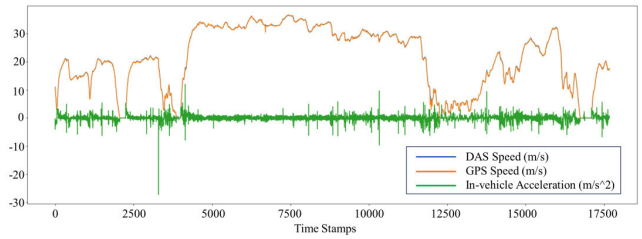


Fig. 1. Snippet of the raw data.

In order to maintain comparability, the CAVs dataset used in [15] and [22] was employed. This dataset was provided by the Safety Pilot Model Deployment (SPMD) program as reported in [34] and can be found in their research data exchange (RDE) database [35]. This dataset was composed of a variety of information generated during the driving process of vehicles, including the Basic Safety Messages (BSM), driving trajectories, driver-vehicle interaction data, and environmental information, collected by the onboard Data Acquisition System (DAS), Global Positioning System (GPS), and roadside units. To keep consistent with [15] and [22], in this work, we employed three sensors to evaluate our proposed sensor anomaly detection method, namely, (1) Sensor 1: speed measured on the vehicle, (2) Sensor 2: speed given by GPS, and (3) Sensor 3: in-vehicle acceleration. A snippet of the raw sensor data is shown in Figrue 1.

It is important to mention here that, in this work, no information about driving control, road conditions, environment, trajectory, or car dynamics was employed, and the external environment of driving was not controlled. Given that a car's driving decisions can be influenced by numerous factors such as traffic conditions, driving behavior, weather conditions, road quality, and legal speed limits, it is highly likely that the statistical properties of the speed of a car will vary over time, making it a non-stationary process which can be expressed by the following equation:

$$y_t = y_{t-1} + u_t, u_t \sim \text{IID}\left(0, \sigma^2\right) \tag{1}$$

where y_t denotes the speed at time t, IID means independent and identically distributed.

Since this dataset did not provide anomalous sensor data, to keep consistent with [15] and [22], the anomaly injection algorithm was kept the same as theirs as shown in Algorithm 1. Four common sensor anomalies caused by cyber-attacks and sensor failures were injected in this work, namely instant, constant, gradual drift, and bias. As stated in [15] and [22], in a practical system, the probability of two sensors being abnormal at the same time can be very low due to the high reliability of the sensors, so we assume that only one sensor can be abnormal at the same time point.

It is important to note here that, in this study, no data normalization operations were employed. This is because, firstly, if sensor anomalies were injected after normalisation, those severe outliers would be significantly different from the normal values and thus make the task less difficult. Secondly, if the data were normalised after anomaly injection, the severe outliers would cause the variance of the original data to become extremely small, thus increasing the demand for the precision of the data processed by the model and resulting in an unnecessary computational burden.

Algorithm 1 Anomaly Injection

$t \leftarrow$ time step
$T \leftarrow$ total number of data points
$p \leftarrow$ anomaly possibility
$i \leftarrow$ sensor index
$n \leftarrow$ number of sensors
$x_i \leftarrow$ sensor data
$x_i' \leftarrow$ sensor anomaly # A full description of the anomaly patterns can be found in [15] and [22]
for $t \in T$ **do**
  if $a \sim U(0,1) \leq p$ **then**
    $i \leftarrow randint(n)$
    $x_i' \leftarrow x_i + Anomaly$
  else
    $x_i' \leftarrow x_i$
  end if
end for

III. Methodology

In this section, an introduction to the overall model architecture of DSA-CNN is provided along with a detailed explanation of the two fundamental building blocks of DSA-CNN, namely DAM block and CNN block. This is followed by an explanation of how the DAM block extracts and integrates the temporal patterns (time-wise attention) and spatial patterns (sensor-wise attention) based on the self-attention mechanism.

A. Overall Architecture of DSA-CNN

The overall architecture can be described by Figure 2. The multiple sensor time series data will be first passed into the class token concatenation function, which will be explained in detail in a later section, to add a class token that will be used as the feature representation for classification. Then, the output of class token concatenation can be fed to the Dual-channel attention mechanism (DAM) block to generate the feature maps of the attention mechanism, followed by a CNN block to further extract features. The DAM block and CNN block will be repeated several times before using a linear layer to perform the final classification. The output of this model can be the anomaly conditions of the data being passed to this model.

In terms of the input, let X denote the multiple sensor streams as shown in the following expression:

$$X = [X_1, X_2, \ldots, X_C] \tag{2}$$

where X_C denotes the C-th sensor data stream. Each sensor stream can be composed of a series of readings over a predefined time window as shown in the following equation:

$$X_C = [x_1, x_2, \ldots, x_L] \tag{3}$$

where L is the number of readings in a time window. Hence the input space of the model can be expressed by $\mathcal{X} \in \mathbb{R}^{C \times L}$.

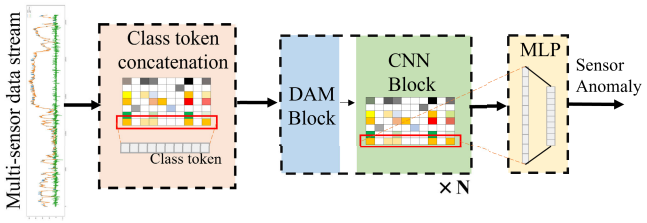


Fig. 2. Architecture of dual-channel self-attention-based convolutional neural network (DSA-CNN) for sensor anomaly detection.

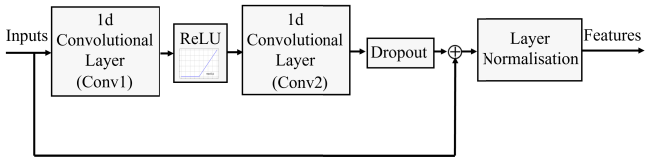


Fig. 3. CNN block.

B. CNN Block

The proposed CNN Blocks employ 1d-CNN as a basic building element, and the architecture can be demonstrated by Figure 3. As shown in Figure 3, the inputs can be first processed by a 1d-CNN layer and fed into a ReLU activation layer, before passing to the second 1d-CNN layer. Then, after the dropout operation, the original inputs can be added to the current feature maps via a shortcut connection. Finally, a normalisation layer is employed to form the final feature maps. The computation flow can be described by the following equation:

$$Out_{CNN} = \mathrm{LN}(\mathrm{Dropout}(\mathrm{Conv2}(\mathrm{ReLU}(\mathrm{Conv1}\,(X)))) + X) \tag{4}$$

where LN denotes the Layer normalisation, $Conv1()$ and $Conv1()$ are the convolution operation, $ReLU()$ is the activation function, X is the input multiple sensor time series.

1) 1d-CNN: CNNs have been proven effective and achieved some of the state-of-the-art results for TSC among data-driven methods in literature [36], [37]. This is because the sensitivity of CNNs to local features can be higher and their feature learning can be spatial invariant, compared with ANNs that treat all features equally [38]. In this study, 1d-CNN was used instead of the normal CNNs with two-dimensional convolutional kernels. This is because the attention mechanism was relied on to model the spatial dependencies, CNNs were only expected to focus on a single series. For using 1d-CNN to TSC, the kernel size can be a key factor affecting performance [39]. Most of the works using 1d-CNN regarded the kernel size as a hyper-parameter and optimised it by grid search which can be time-consuming and computation-expensive. As this work does not rely on 1d-CNN to capture the major spatiotemporal features, a small kernel of size (1, 3) was used. This is because, firstly, smaller kernels can be computationally efficient. Secondly, as the speed was assumed non-stationary, which means the signal behaviour is likely to vary over a long period, and therefore the feature in a smaller perception field can be more representative.

In the proposed CNN blocks, the dimensionality of the output feature map can be kept equal to the dimensionality of the input data to maintain extensibility. This can be achieved by: (1) organising the number of output channels of the second convolutional layer to be equal to the number of input channels of the first convolutional layer, (2) configuring the convolutional kernels to 3 in size with a stride of 1 and padding of 1. This means the length of input and output can be kept the same as shown in the following equation:

$$L_{out} = \frac{L_{in} + 2 \times padding - (kernel_size - 1) - 1}{stride} + 1 \tag{5}$$

2) Activation Layer: As the convolution operation can be considered as a linear mapping, a non-linear activation layer is used after the first 1d convolution layer to introduce non-linear modelling capabilities and thus enhance the representativeness of the learned features, enhancing the capacity to model complex input space. In the CNN Blocks of our proposed model, the ReLU is employed as the activation function, which can be expressed by the following equation:

$$\mathrm{ReLU}(H) = \max(0, H) \tag{6}$$

where H is the feature map passed to ReLU. This is because ReLU has a significant advantage over other activation functions such as tanh and sigmoid in terms of computational complexity, mitigation of the gradient saturation problem, and sparsity [40].

3) Layer Normalisation: The normalisation layer can be a critical component in the deep neural network. By normalising the inputs to a layer to zero mean and unit variance, the training process can be stabilised and the performance of the network can also be improved [41], [42]. Batch Normalisation (BN) and Layer Normalisation (LN) can be the prevailing normalisation methods. While BN normalises each feature within a sample, LN normalises all features within each sample. To be specific, BN erases the relative magnitudes of the different features but preserves the relative magnitudes of the different samples. Hence, BN can be more effective when the statistical behaviours among different samples are more important. In contrast, LN wipes out the relative magnitudes between different samples but keeps the relative magnitudes between different features within a sample. Therefore, it can be more suitable for tasks where the features within a sample are closely related, such as anomaly detection for multiple sensors. Speaking to the TSC task on CAV with unmanaged driving behaviour, as the speeds and acceleration can be treated as non-stationary processes, the statistical features among samples can be very uninformative for sensor anomaly detection. As a result, LN can be more suitable for this task.

LN performs normalisation over the last 2 dimensions based on the following equation:

$$y = \frac{x - \mathrm{E}[x]}{\sqrt{\mathrm{Var}[x] + \epsilon}} * \gamma + \beta \tag{7}$$

where y and x denote the normalised matrix and input matrix respectively, ϵ is a minimum preventing the denominator to be zero, γ and β are trainable affine transform weight and

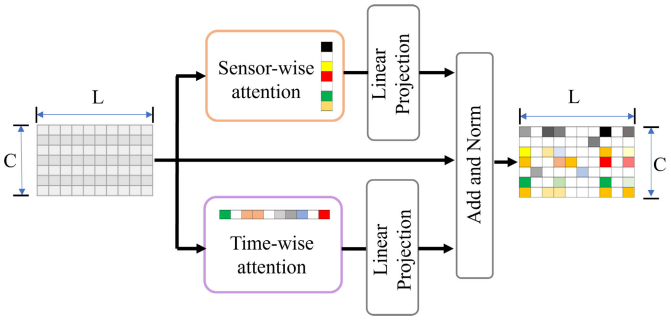


Fig. 4. Dual-channel attention mechanism (DAM).

bias. E[x] and Var[x] denote the mean and variance of input x, which can be defined by the following equations:

$$E(x) = (1/t)\sum_{i=1}^{t} x_i \tag{8}$$

$$\mathrm{Var}[x] = (1/t)\sum_{i=1}^{t} (x_i - E(x))^2 \tag{9}$$

C. Dual-Channel Attention Mechanism (DAM)

In multiple sensor anomaly detection tasks, abnormal data can appear at any time and in any sensor, making both temporal and spatial (sensor-wise) features very important. In this work, the Dual-channel Attention Mechanism (DAM) was proposed to integrate the learning of spatiotemporal features into the training process, hence the spatiotemporal features can be extracted progressively and automatically.

The overall architecture of DAM can be illustrated by Figure 4. The input can be passed to two attention modules simultaneously, namely sensor-wise attention and time-wise attention. The former is responsible for extracting spatial dependencies among different sensors and the latter is for extracting temporal dependencies among different time points. Then, the feature maps generated by these two modules and the original input will be added together, forming a shortcut connection, before normalisation. The output of the DAM is therefore a feature map with integrated spatiotemporal features. The computation flow of DAM can be described by the following equation:

$$OUT_{DAM} = \mathrm{LN}(\mathrm{LP}(\mathrm{SWA}(X)) + \mathrm{LP}(\mathrm{TWA}(X^T)^T) + X) \tag{10}$$

where LP denotes linear projection, SWA and TWA mean sensor-wise attention and time-wise attention respectively. The details of sensor-wise attention and time-wise attention will be explained in the following section.

1) Sensor-Wise Attention: The sensor-wise attention processes the dependencies among different sensors based on the self-attention mechanism proposed in [26] as shown in Figure 5a. Analogous to the operation of retrieval systems, the self-attention mechanism utilizes a query vector to initiate an information search. The mechanism subsequently identifies corresponding keys in its database to align with the query vector, ultimately resulting in an output represented by value

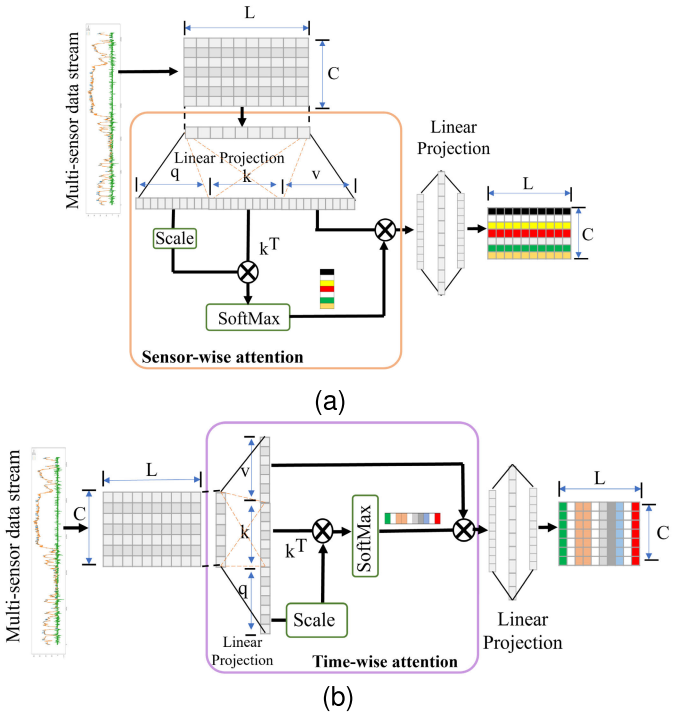


Fig. 5. (a) Sensor-wise attention. (b) Time-wise attention.

vectors linked to these keys. In the context of the self-attention framework, input sequences can be converted via linear projection into query vectors (Q), key vectors (K), and value vectors (V), which is visually represented in Figure 5a and elaborated on in the following equations:

$$Y = XW^T + B \tag{11}$$

where $X \in \mathbb{R}^{C\times L}$, $W \in \mathbb{R}^{3L\times L}$, $Y, B \in \mathbb{R}^{C\times 3L}$, and

$$Q, K, V = Y[0:C, 0:L], Y[0:C, L:2L], Y[0:C, 2L:3L] \tag{12}$$

where $Y[a:b, c:d]$ denotes the a to b rows and c to d columns of Y. Then, the attention weights can be calculated by:

$$Attention_weight = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right) \tag{13}$$

where d_k represents the dimension of K and $1/\sqrt{d_k}$ is a scalar to prevent the predominance of a particular term in SoftMax calculation, which might challenge the calculation of the gradient [26]. Given that Q and K are independent and adhere to the Gaussian distribution, the variance of their dot product will be d_k, posing a challenge when dealing with the data with high dimensionality [26]. Finally, the attention weight is multiplied by the V to form the output feature map of self-attention:

$$Attention\,(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{14}$$

Clearly, the attention weights regulate the information transfer. If a certain segment of the input data is assigned a weight of zero, the propagation of information to the subsequent network

layer can be impeded. Furthermore, each element (row) within the attention output sequence integrates the information of all elements in the entire input sequence. Hence, the dependencies among different sensors can be modelled and integrated. It can be found from the above calculations that, the dimension of the output feature maps remains the same as the dimension of the input data, namely $Attention\,(Q, K, V) \in \mathbb{R}^{C\times L}$.

2) Time-Wise Attention: Similarly, in time-wise attention, the base unit for carrying out the calculation of the attention mechanism can be replaced by the time steps, as shown in Figure 5. Time-wise attention can be simply achieved by transposing the input and then performing similar computations to obtain the Q, K and V before calculating the attention weights:

$$Y = XW^T + B \tag{15}$$

where $X \in \mathbb{R}^{L\times C}$, $W \in \mathbb{R}^{3C\times C}$, $Y, B \in \mathbb{R}^{L\times 3C}$, and

$$Q, K, V = Y[0:L, 0:C], Y[0:L, C:2C], Y[0:L, 2C:3C] \tag{16}$$

Therefore, the feature maps with the integrated time dependencies over the entire time window can be obtained.

Then, the feature maps of Time-wise attention can be transposed to keep the equal dimensionality with Sensor-wise attention, before adding them to the original input to make a shortcut connection. Finally, the constructed feature map can be normalised, forming the output feature map of the DAM block as shown in Figure 4.

D. Class Token Concatenation

Based on equation 14, it can be found that each row of the output of the attention mechanism (corresponding to each sensor in this work) is a weighted sum of all rows of the feature map from the previous layer, and the weights are calculated based on the data from each sensor. Theoretically, it is reasonable to take any row in the final output of the model's feature map and use it as a multi-sensor feature representation to perform classification. However, to avoid the influence of specific sensor data on the feature representation of multi-sensor data, a vector of random numbers was added to the input data as a token for classification [26]. The features of all sensors can be weighted and integrated into this token, hence this vector can be used as a final feature representation. This operation is illustrated in Figure 2, and the details can be found in algorithm 2.

The overall computation flow of the proposed DSA-CNN can be illustrated by algorithm 2.

Algorithm 2 Dual-Channel Attention CNN (DSA-CNN)

Input: $X \leftarrow$ CAV Sensor Readings
Output: Normal reading, Anomalous reading
$X \leftarrow X + Anomalies$ # Anomaly injection
Model Installation:
for l in range(Number of layers in DSA-CNN) **do**
  Initiate $Weights^{(l)} \sim N(0, 1)$
  Perform Singular Value Decomposition to $Weights^{(l)}$
    Ensure: $Weights^{(l)} Weights^{(l)T} = I$
Class Token Concatenation
$Token \leftarrow rand((1, L))$
$X \leftarrow Concatenate((X, Token), dim = 0)$
Forward:
for i in range(N) **do**
  # DAM block calculations
  $X_{SWA} \leftarrow LP(SWA(X))$
  $X_{TWA} \leftarrow LP(TWA(X^T)^T)$
  $X \leftarrow X_{SWA} + X_{TWA} + X$
  $X \leftarrow LN(X)$
  # CNN block calculations
  $Residual \leftarrow X$
  $X \leftarrow ReLU(Conv1(X))$ # Conv1
  $X \leftarrow Dropout(Conv2(X))$ # Conv2
  $X \leftarrow LN(X + Residual)$
$Outputs = \text{Classifier}(X) = XW_{classifier} + b_{classifier}$
Training:
for epoch in range(Number of epochs) **do**
  Calculate Loss
  Calculate Accuracy, Precision, Sensitivity, F-Score
  Backpropagation
Return: Output

IV. Experiments and Results

A. Training Process

This experiment was conducted on Google Colab environment with NVIDIA Tesla P100 PCIe 16 GB, and PyTorch 1.12 was used as the deep learning framework. The hyperparameters of the proposed model can be found in the Appendix. As we formulated the sensor reading anomaly detection task as a classification task, the cross entropy loss was used as the loss function which can be defined by the following equation:

$$\text{Loss}(\text{y}, \hat{\text{y}}) = -\sum_{\text{i}=1}^{\text{C}} \text{y}_\text{i} \log\left(\hat{\text{y}}_\text{i}\right) \tag{17}$$

where y and $\hat{\text{y}}$ are the true labels and model prediction using one-hot encoding.

In terms of model initialisation, orthogonal initialisation was employed, namely, the weight matrix $W^{(l)}$ of each layer will be initialised to an orthogonal matrix, satisfying:

$$W^{(l)} W^{(l)^T} = I \tag{18}$$

where I is an identity matrix, and l is the layer index. This is because the orthogonal initialisation is able to make the error term norm-preserving during the backpropagation process [43], which can be expressed in the following equation:

$$\left\|\delta^{(l-1)}\right\|^2 = \left\|W^{(l)^T}\delta^{(l)}\right\|^2 = \left\|\delta^{(l)}\right\|^2 \tag{19}$$

where $\delta^{(l)}$ is the loss term of the l layer of the model, and it has shown great efficiency practically for training the model based on the attention mechanism [44], [45]. Since in our proposed model, the activation function was all ReLU and the average gradient of ReLU around 0 can be approximated

as 0.5, we multiplied the initialised weight matrix by $\sqrt{2}$ to preserve norm-preserving property. This is denoted by $Gain = 1.41$ in Table VI.

B. Results Evaluation Method

The performance of the proposed method was evaluated by the following scores which can be normally used for assessing classification algorithms:

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN} \tag{20}$$

$$\text{Sensitivity} = \frac{TP}{TP + FN} \tag{21}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{22}$$

$$\text{F1 Score} = \frac{2(\text{ Sensitivity } * \text{ Precision })}{\text{Sensitivity } + \text{ Precision}} \tag{23}$$

where TP, TN, FP, and FN mean True Positive, True Negative, False Positive and False Negative respectively. In terms of the sensor anomalies detection domain, as undetected abnormalities are far more damaging than classifying normal readings as abnormal, the sensitivity score was highlighted in the results. As the F1 score can be a comprehensive measure of precision and sensitivity, it was used as a reference to compare the performance of different models. Additional evaluations, such as the ROC curve and the FPR95 metric (the false positive rate at the true positive rate equal to 95%), which can be critical in anomaly detection tasks can be found in the Appendix.

C. Single Anomaly Detection

In this section, the performance of DSA-CNN is evaluated with respect to the occurrence of a single type of anomaly. To ensure comparability, the same anomaly pattern, severity, and possibility (5%) were used as in the current state of the art [15] and [22]. The results from CNN Kalman Filter (CNN-KF) [15], Multi-stage attention mechanism with an LSTM-based CNN (MSALSTM-CNN) [22], and our proposed DSA-CNN, in instant, bias, constant and drift detection are compared in Table I to IV respectively. The mean F1 score and sensitivity comparison for each type of anomaly is summarised in Figure 6.

1) Instant Anomaly Detection: Table I shows the performance of DSA-CNN compared with the methods working on the same dataset in literature for instant anomaly detection. As the severity of the anomaly increases, an improvement in the performance of all methods can be observed. However, the proposed DSA-CNN achieves higher F1 scores for all severity levels of instant anomaly, compared with the SOTA performance achieved by MSALSTM-CNN in the literature [22]. Speaking to sensitivity, except for the magnitude of $500 \times N(0, 0.01)$, DSA-CNN can be also better than the other methods significantly. It is worth noting that DSA-CNN shows outstanding performance in detecting mild anomalies. While the F1 score of the MSALSTM-CNN at severity $25 \times N(0, 0.01)$ is 70.18%, the DSA-CNN achieves 78.82%, and the sensitivity at this severity can also be improved by 16.65% by DSA-CNN. In case of severe anomalies, DSA-CNN shows a slight improvement.

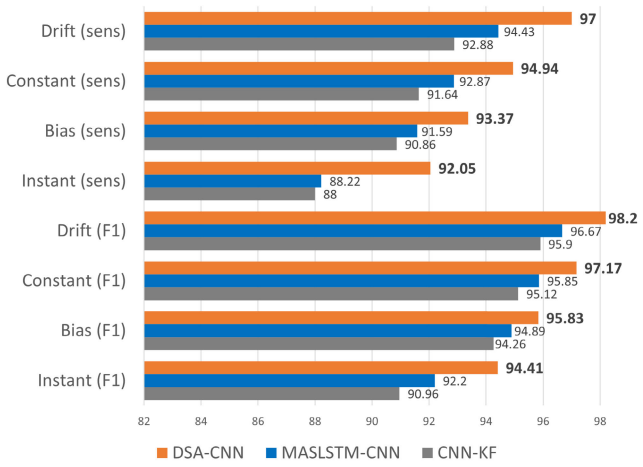


Fig. 6. Performance comparison of different models for single anomaly detection.

2) Bias Anomaly Detection: Table II shows the performance comparison for bias anomaly detection. For all methods in this table, the longer the duration of the anomaly, the higher the detection performance of the models for the same magnitude. Similarly, for anomalies of the same duration, the higher the magnitude of the anomaly, the higher the detection performance. Based on the results in this table, our proposed DSA-CNN achieves improvements in both sensitivity and F1 scores among most of the severity, compared to SOTA performance in the literature, except for the magnitude of $U(0, 1)$ with a duration of 3, where MSALSTM-CNN achieves the best F1 score (90.57%) and ours is 90.38%. However, it can be found that DSA-CNN has significant advantages in anomaly detection of small duration. In anomalies of duration 3 and 5, DSA-CNN improved on average by 2.57% and 1.28% in sensitivity and F1 score respectively compared to MSALSTM-CNN.

3) Constant Anomaly Detection: Table III shows the performance comparison for Constant anomaly detection. It can be observed that the performance of CNN-KF, MSALSTM-CNN, and DSA-CNN methods consistently improves as the anomaly duration increases while keeping the anomaly magnitude constant (U(0,5)). Their performance decreases as the anomaly magnitude reduces while keeping the duration constant (d=10). The DSA-CNN method, however, demonstrates superior performance in terms of sensitivity and F1 score across all experiments. In addition, our proposed method displays consistency in its performance, whereas the CNN-KF and MSALSTM-CNN methods exhibit some variation. Nevertheless, all three methods show slightly declining performance in the last row with the lowest anomaly magnitude (U(0,1)).

4) Drift Anomaly Detection: Table IV shows the results of gradual drift anomaly detection using the CNN-KF, MSALSTM-CNN, and DSA-CNN models. The MSALSTM-CNN and DSA-CNN methods exhibit satisfactory performance, while DSA-CNN outperforms MSALSTM-CNN in sensitivity and F1 score for all experiments. The highest F1-score of 98.97% is observed in row 2 by the DSA-CNN

TABLE I
INSTANT

	CNN-KF(%) [15]				MSALSTM-CNN(%) [22]				DSA-CNN(%)			
Anomaly Magnitude	Acc	Sens	Prec	F1	Acc	Sens	Prec	F1	Acc	Sens	Prec	F1
25 x N(0,0.01)	80.0	51.5	97.6	67.4	84.10	54.63	98.12	70.18	85.00	**71.28**	88.16	**78.82**
100 x N(0,0.01)	93.6	86.2	97.9	91.7	95.8	89.60	98.43	93.80	95.42	**91.00**	97.85	**94.30**
500 x N(0,0.01)	98.3	96.0	99.7	97.8	96.02	**99.79**	97.86	99.02	99.58	99.00	99.99	**99.50**
1000 x N(0,0.01)	98.8	97.1	99.8	98.4	98.98	98.16	99.21	98.68	99.58	**99.05**	99.99	**99.52**
10000 x N(0,0.01)	99.7	99.2	99.8	99.5	99.43	98.93	99.75	99.34	99.91	**99.92**	99.92	**99.92**

TABLE II
BIAS

		CNN-KF(%) [15]				MSALSTM-CNN(%) [22]				DSA-CNN(%)			
Anomaly Magnitude	Duration, d	Acc	Sens	Prec	F1	Acc	Sens	Prec	F1	Acc	Sens	Prec	F1
U(0,5)	3	94.6	89.3	99.5	94.2	95.10	90.53	99.47	94.78	95.92	**93.86**	98.79	**96.26**
U(0,5)	5	94.8	90.6	99.6	94.9	95.51	91.99	99.55	95.62	96.13	**93.80**	99.76	**96.69**
U(0,5)	10	95.9	94.4	99.1	96.7	96.56	95.74	99.06	97.37	97.12	**96.69**	99.05	**97.84**
U(0,3)	10	94.4	95.6	98.6	95.5	94.99	93.82	98.50	96.10	97.55	**96.24**	99.73	**97.95**
U(0,1)	10	88.0	84.8	95.9	90.0	88.55	85.81	95.89	**90.57**	88.69	**86.18**	95.02	90.38

TABLE III
CONSTANT

		CNN-KF(%) [15]				MSALSTM-CNN(%) [22]				DSA-CNN(%)			
Anomaly Magnitude	Duration, d	Acc	Sens	Prec	F1	Acc	Sens	Prec	F1	Acc	Sens	Prec	F1
U(0,5)	3	94.9	89.9	99.6	94.5	95.05	90.25	99.51	94.65	95.73	**92.71**	99.04	**95.77**
U(0,5)	5	95.1	91.7	99.0	95.2	95.40	92.30	98.97	95.51	96.98	**94.15**	99.66	**96.83**
U(0,5)	10	96.2	94.9	99.1	97.0	96.61	95.61	99.28	97.41	98.59	**97.71**	99.59	**98.64**
U(0,3)	10	95.3	93.9	98.7	96.2	96.44	95.44	98.93	97.15	98.33	**97.18**	99.59	**98.37**
U(0,1)	10	91.2	87.8	98.6	92.7	93.02	90.76	98.69	94.55	96.56	**92.97**	99.76	**96.25**

TABLE IV
DRIFT

		CNN-KF(%) [15]				MSALSTM-CNN(%) [22]				DSA-CNN(%)			
Anomaly Magnitude	Duration, d	Acc	Sens	Prec	F1	Acc	Sens	Prec	F1	Acc	Sens	Prec	F1
Linespace(0,4)	10	94.7	93.1	99.2	96.1	96.01	95.85	99.14	97.46	98.59	**96.99**	99.61	**98.29**
Linespace(0,4)	20	96.0	95.6	99.3	97.4	96.21	96.03	99.27	97.62	99.11	**98.67**	99.27	**98.97**
Linespace(0,2)	10	93.0	89.6	99.3	94.2	94.36	93.09	99.07	95.58	98.39	**97.32**	99.47	**98.38**
Linespace(0,2)	20	93.7	93.2	98.7	95.9	94.09	92.78	99.52	96.03	97.34	**94.98**	99.43	**97.15**

method when the duration is 20 and the anomaly magnitude is linespace(0,4). The lowest F1-score is observed in row 3, with 94.2% achieved by the CNN-KF method when the anomaly magnitude is the smallest. Furthermore, compared with the MSALSTM-CNN and CNN-KF, the DSA-CNN demonstrates consistent performance throughout the experiments, with minor fluctuations. In [15], Wyk et al. stated that detecting gradual drift can be one of the most challenging tasks of anomaly detection. In the hardest situation, namely the magnitude of linespace(0,2) with a duration of 20, DSA-CNN can still improve the sensitivity and F1 score by 2.2% and 1.12% respectively.

D. Mixed Anomaly Detection

In our experiment, the performance of DSA-CNN was also examined for the occurrences of all types of anomalies. The performance evaluation of different algorithms with respect to different sensors is given in Table V. As this type of experiment is not provided in [22], only [15] and our proposed method are shown in this table. Based on the results, DSA-CNN outperforms the other algorithms on all three sensors, achieving the highest scores on almost all performance metrics, especially for sensitivity and F1 scores. This suggests that DSA-CNN provides a more accurate and robust approach to anomaly detection than the other algorithms evaluated.

TABLE V
MIX

	KF(%) [15]				CNN(%) [15]				CNN-KF(%) [15]				DSA-CNN(%)			
Sensor	Acc	Sens	Prec	F1	Acc	Sens	Prec	F1	Acc	Sens	Prec	F1	Acc	Sens	Prec	F1
1	97.8	90.5	95.9	93.1	98.0	93.3	99.5	96.3	98.1	94.2	99.4	96.7	99.32	**98.42**	98.17	**98.30**
2	98.0	90.6	96.0	93.2	97.0	90.4	98.7	94.4	97.6	92.9	98.4	95.5	98.70	**96.12**	98.14	**97.12**
3	96.1	85.1	89.0	86.9	94.7	83.3	97.6	89.8	95.7	87.3	97.3	92.0	95.57	**89.36**	97.11	**93.07**

V. DISCUSSION

It is clear that the DSA-CNN proposed in this paper offers significant advantages in terms of sensitivity and F1 score, especially in minor sensor anomalies. Higher sensitivity means fewer undetected anomalies, which can be crucial for safety-critical application scenarios, such as the CAV velocity and acceleration data used in this work, which can be the important decision bases for autonomous driving systems [46].

As an important sensor anomaly detection algorithm, Kalman filtering has the advantages of being easy to implement and computationally efficient, making it widely used. It can predict the near future or current system state by tracing back to a few past system state data, so by comparing the difference between the estimated data and the measurements and setting a threshold, it can determine if the sensor reading now is in an abnormal state. However, Kalman filtering has three disadvantages in sensor anomaly detection. Firstly, its performance relies on artificial noise estimations, for example in [15] where the sensor noise was assumed to be Gaussian white noise, which is sometimes not representative of the real noise in the actual system. Secondly, plain Kalman filtering often requires the assumption that the system is linear and time-invariant [47]. This assumption can be satisfied when the time window taken for the analysis of CAV velocities is small enough, but the time window for the analysis of time-series data often cannot be too small to ensure that sufficient information can be incorporated, so the non-linearity introduced by the large time window becomes a source of error in the Kalman filter estimations. Finally, defining thresholds for normal intervals can be often difficult, as the intensity of noise in a real system can be likely to vary according to environmental conditions. These might be the reason why the Kalman filter-based approach in [15] achieves relatively poor performance despite the fact that the use of CNNs to extract a representation of the system state can overcome to some extent the problems of noise and non-linearity within the time window. Since deep learning-based methods do not require the aforementioned assumptions, they offer great potential for anomaly detection due to their fewer restrictions on applications and enhanced applicability.

LSTM is an RNN algorithm that excels in time series data modelling, and plays an important role in the detection of anomalies in time-series sensor data. LSTM-based methods do not require an accurate estimation of noise, unlike Kalman filtering, and do not rely on the assumption of linear time-invariant systems. This can be the possible reason why the LSTM-based method used in [22] outperforms the Kalman filter-based method in [15]. However, since LSTM can be still recursive in nature, the features of the initial time steps can still be difficult to learn, especially for small anomalies. In terms of the proposed DSA-CNN, it combines the strengths of CNNs in local feature extraction with the self-attention mechanism's ability to model global context. This synergy enhances the model's sensitivity to small anomalies in multi-sensor data. Traditional methods might overlook these small anomalies due to their reliance on local patterns and inability to capture long-range dependencies effectively. In addition, the integration of a dual-channel self-attention mechanism allows for the extraction of spatiotemporal features from sensor data, enabling the system to extract subtle but crucial details that might indicate an anomaly. This is particularly important for small anomalies that may not produce significant changes in sensor readings. The DSA-CNN method leverages these

TABLE VI
HYPER PARAMETERS

CNN Block	
Conv1	channel_in = 15, channel_out=60, kernel size = 3, padding = 1, bias=False
Conv2	channel_in = 60, channel_out=45, kernel size = 3, padding = 1, bias=False
LN	ϵ = 1e-6
Dropout	0.2
SWA	
Self-attention	qkv dimension (L) = 15
Linear projection	Dropout = 0.1, in = 15, out = 15
TWA	
Self-attention	qkv dimension (L) = 3
Linear projection	Dropout = 0.1, in = 3, out = 3
Classifier	
Linear layer	in = 15, out = number of classes
Training	
Model depth (N)	7 (Number of DA-CNN blocks)
Initialisation	Orthogonal, Gain = 1.41
Batch size	16
Optimiser	Adam
Learning rate	0.0001
Epoch	300

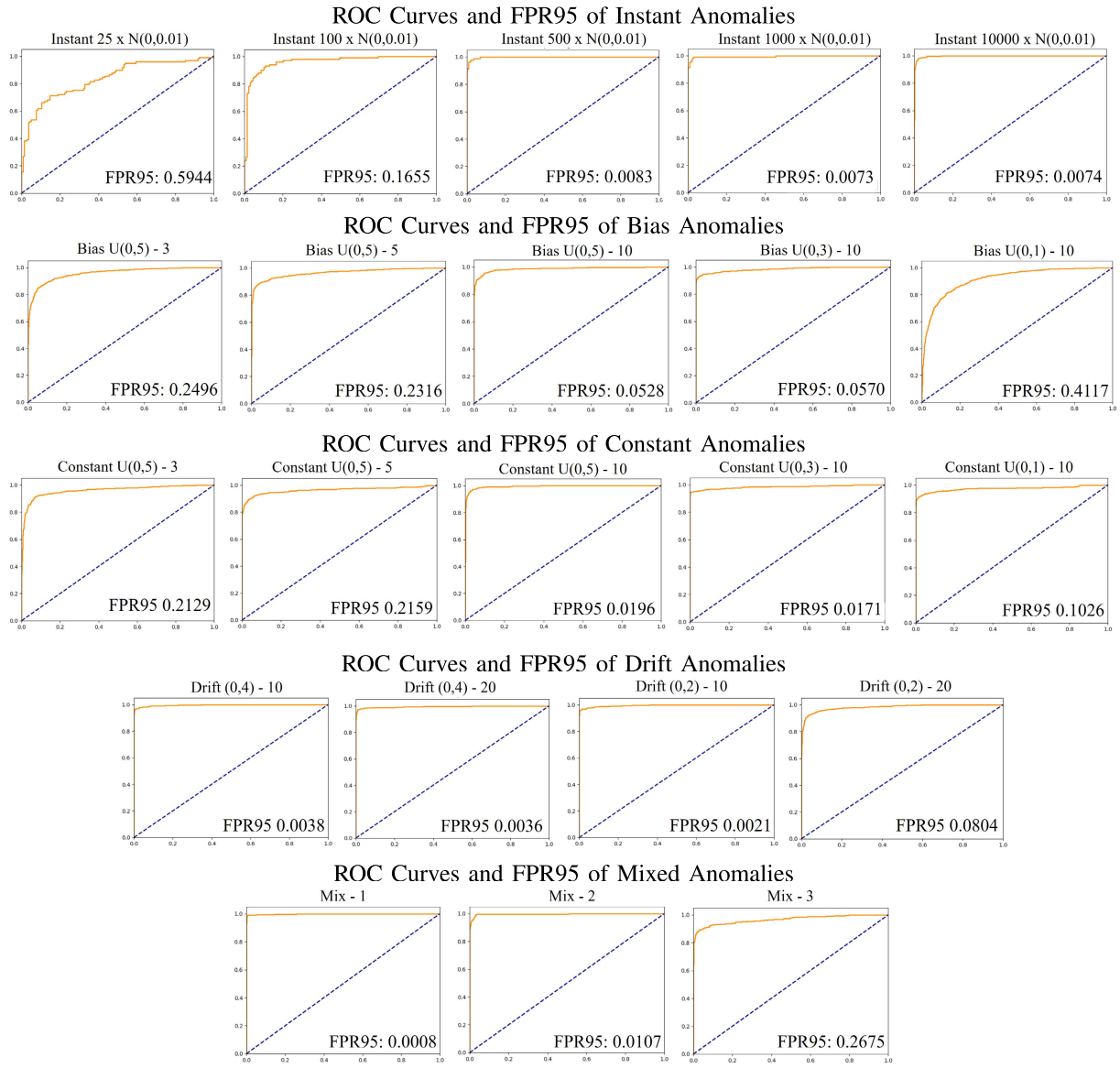


Fig. 7. ROC curves and FPR95 metrics of the experiments conducted in this study.

advancements, offering a more robust and sensitive approach to anomaly detection. Additional experiments to verify the effectiveness of the proposed dual-channel self-attention mechanism can be found in the Appendix.

However, due to the current supervised learning-based training approach, the performance of the proposed model in dealing with unknown complex anomalous patterns has not been evaluated, which can be important in real applications [48]. A common deep learning-based approach in the literature for handling anomalies of unknown patterns can be the unsupervised learning-based autoencoder architecture. In this type of approach, the original normal signal can be mapped to latent spaces using a deep learning model, and then, one of the latent spaces can be used as a feature representation. The distance between the latent space of the anomalous data and the latent space of the normal data can be calculated as a criterion for the occurrence of anomalies [49], [50], [51]. It can be found that the patterns of anomaly behaviour will not be treated as prior knowledge, giving this type of method the potential to detect arbitrary anomalies. The performance of DSA-CNN in this type of methodology can be an important future direction to be evaluated. In addition, the assumption of anomaly that only one sensor can be abnormal at the same time can be violated in real-world applications. This setting was adopted to maintain consistency with the two baselines referenced in this study, thereby enabling the comparison of algorithmic performance. Extending the proposed approach to scenarios with multiple simultaneous sensor anomalies can be another valuable direction for further research.

VI. Conclusion and Future Work

Anomaly detection in multi-sensor data streams can be an important foundation for CAVs and many safety-critical systems. In this paper, a method, DSA-CNN, was proposed based on the self-attention mechanism and CNN to detect the anomalous behaviour of CAV velocity and acceleration

TABLE VII
PERFORMANCE DIFFERENCE AMONG DSA-CNN, ITS SENSOR-WISE ATTENTION-ONLY VERSION, AND ITS TIME-WISE ATTENTION-ONLY VERSION

Tasks	DSA-CNN				Sensor-wise attention only				Time-wise attention only			
	Acc	Sen	Prec	F1	Acc	Sen	Prec	F1	Acc	Sen	Prec	F1
Instant 25 x N(0,0.01)	85.00	**71.28**	88.16	**78.82**	82.50	70.70	84.33	76.92	63.75	44.90	57.14	50.29
Bias U(0,1)-10	88.69	**86.18**	95.02	**90.38**	84.13	83.13	90.35	86.59	71.33	76.83	76.92	76.88
Constant U(0,5)-3	95.73	**92.72**	99.04	**95.77**	95.65	92.13	99.48	95.67	92.23	88.31	96.37	92.17
Drift Linespace(0,2)-20	97.34	**94.98**	99.43	**97.15**	96.56	93.48	99.30	96.30	88.07	84.95	89.71	87.27
Mix - 3	95.57	89.36	97.11	**93.07**	94.58	**89.89**	93.49	91.70	92.23	85.58	90.78	88.10

sensors. We proposed a dual-channel self-attention structure to enhance the sensitivity of the model with respect to small anomalies, and this structure can integrate the learning of spatiotemporal features into the training process of deep learning, eliminating artificial feature engineering. Our proposed method was tested on a publicly available CAV dataset, and the sensor anomaly was simulated based on the 4 most damaging anomalies for CAVs [32], [33]. Based on the experimental results, our proposed DSA-CNN achieved 2.57%, 2.07%, 1.78%, and 3.83% performance gains in sensitivity for drift, constant, bias, and instant anomalies, and 1.53%, 1.32%, 0.94%, and 2.21% gains in F1 score, compared with the SOTA performance model in the literature proposed in [22]. The experimental results also show that DSA-CNN has a clear advantage in handling small anomalies.

However, due to the current supervised-learning-based classification training scheme, it can be difficult to evaluate the performance in the presence of unknown anomaly patterns in a convincing manner. By reorganizing the deep learning model proposed in this paper into an unsupervised learning form, all sensor data can be mapped to a latent space. By comparing the distance of anomalous data to normal data in the latent space, the presence of sensor anomaly can be determined. As a result, this type of model can be no longer limited by specific anomaly patterns, improving the generalisation ability. This can be an important future research direction.

APPENDIX
HYPER PARAMETERS OF DSA-CNN

The hyperparameters of the model proposed in this work are shown in Table VI.

ADDITIONAL EVALUATIONS

This section provides additional evaluations, namely the ROC curve and the FPR95 metric (the false positive rate at the true positive rate equal to 95%), which can be critical in anomaly detection tasks. The ROC and FPR95 can be found in Figure 7.

ABLATION EXPERIMENT

In order to verify the positive effect of one of the core modules of the model proposed in this paper, namely the DAM module, on performance, this section presents ablation experiments.

Two distinct experiments were conducted to validate the efficacy of integrating sensor-wise and time-wise attention mechanisms. In the initial experiment, we exclusively employed sensor-wise attention across both channels, focusing on the individual characteristics of each sensor. Conversely, the second experiment solely utilized time-wise attention in both channels, concentrating on temporal dynamics. This ablation study was designed to rigorously assess the impact and effectiveness of combining sensor-wise and time-wise attention, providing a comprehensive understanding of their synergistic potential in the proposed model.

To maintain comparability, all training parameters, data, the rest of the model architecture, as well as the environment were kept the same. The ablation experiments were only conducted on the most challenging tasks on each of the anomaly types (The tasks that received the lowest F1 scores). The results can be found in Table VII. It can be found that neither the use of sensor-wise attention alone nor time-wise attention can achieve the performance of using them simultaneously, which validates the effectiveness of the proposed DAM in anomaly detection.

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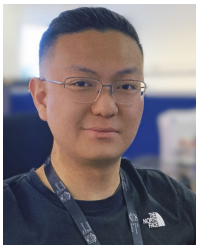
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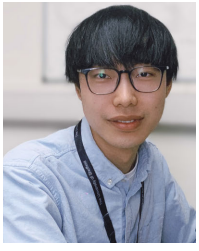
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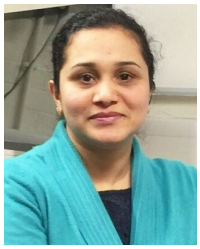
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